

25COP928

Professionalism, Ethics and Cyber Security

Introduction

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Today's Session Structure

- **Module Organization**
 - Teaching Schedule
 - Overview of Lecture/Seminar Topics
 - Assessment Overview
- **Codes of Conduct and Ethics**
- **Ethical Theories**
- **Professional Codes of Ethics**



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Teaching Schedule

Session	Day	Time	Weeks	Room
Lecture	Monday	14:00–15:00	Weeks 1–3, 5–10	U005
Lecture	Thursday	13:00–14:00	Week 4	U020
Seminar (Group 1)	Friday	15:00–17:00	Week 1	U005
Seminar (Group 1)	Monday	15:00–17:00	Weeks 2–10	U005
Seminar (Group 2)	Friday	13:00–15:00	Weeks 1–9	U005
Seminar (Group 2)	Thursday	13:00–15:00	Week 10	U005

Overview of Lecture Topics

- **Codes of Conduct**
- **Data Protection and Data Processing**
- **Intellectual Property and Code**
- **Information Governance**
- **International Digital Issues I: Cyber War**
- **Cyber Security and Ethical Hacking: Practices and Considerations**
- **Cyber Security and the Law (UK)**
- **International Digital Issues II: Supply Chains and Import/Export**
- **Workplace and Employment: Professionalism in Teams**
- **Workplace and Employment: Whistleblowing and Morality**



Overview of Seminar Topics

- **Assessing and designing codes of conduct**
- ****Identifying special data categories****
- **Describing and protecting IP**
- ****Identifying standards for Information Governance Challenges****
- **Hybrid war: The levers of state**
- **Ethics for cyber activities: LEON and accountability frameworks**
- **Legal advice / legal awareness in incident response**
- **Hybrid War: Non-state actors and attribution**
- **Dealing with difficult colleagues**
- **Moral decision making**



Assessment Overview

Main assessment

100% Exam

Exception

Part-time students are assessed by 100% coursework

The assessment is based on content from:

- Lectures
- Seminars
- All topics covered across the module

Teaching material

Lectures

- Lecture notes uploaded **before the lecture**
- Lecture captures available **after the lecture**

Seminars

- Seminar materials uploaded **before the session**
- Solutions and discussion points shared **after the seminar**

Additional Resources

- Research papers
- Further reading and useful links

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Codes of conduct and ethics

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What is common between them ?



Frances Haugen



- **Frances Haugen** has an academic background in electrical and computer engineering.
- She is a specialist in **algorithm management** and worked in several major tech companies, focusing on how algorithms sort and recommend information on platforms such as **Google, Pinterest, Yelp, and Facebook**.
- She later shared confidential documents that suggested Facebook prioritised profit over user protection, which contributed to the spread of misinformation and negatively affected young people's mental health.



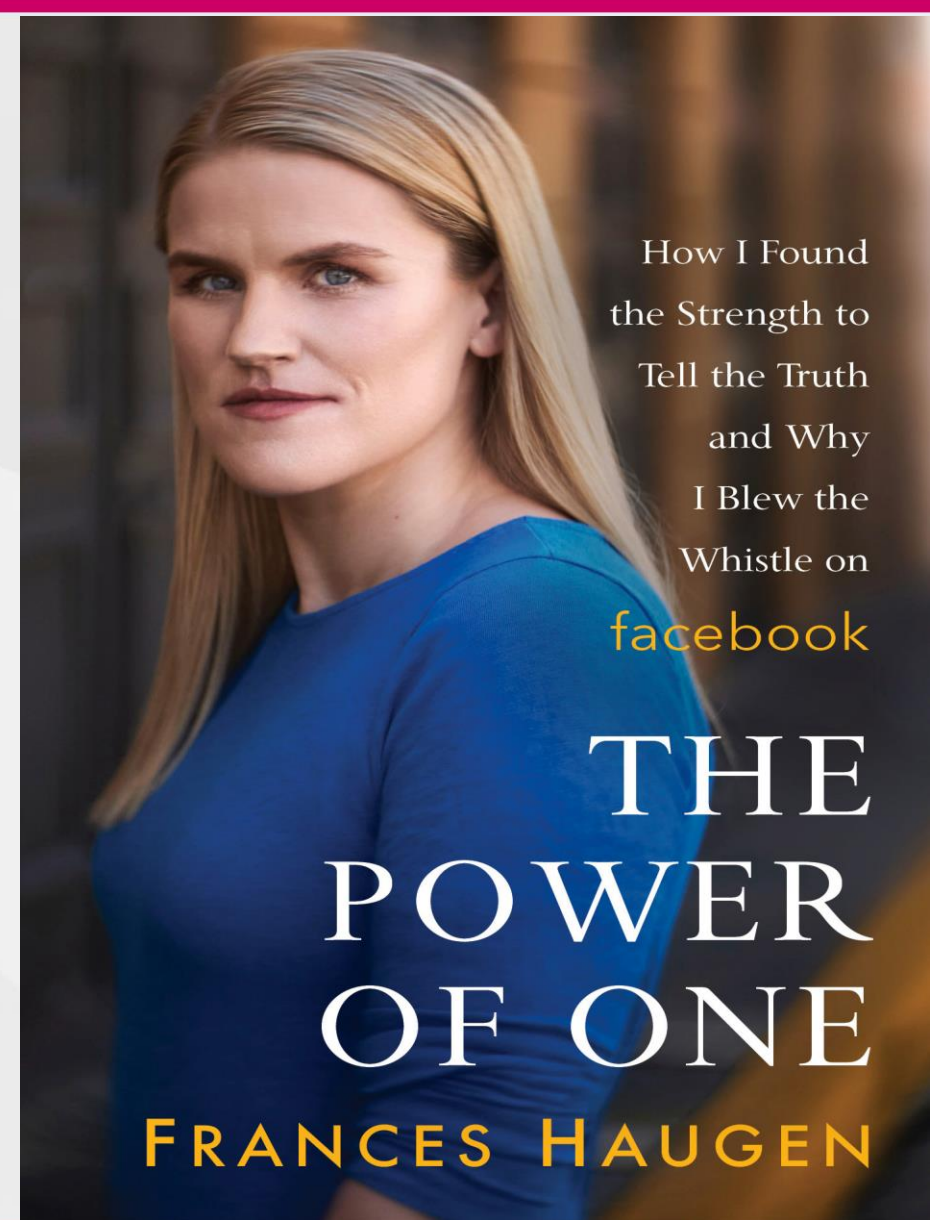
Frances Haugen

Frances said:

"The thing I saw at Facebook over and over again was there were conflicts of interest between **what was good for the public and what was good for Facebook**. And Facebook, over and over again, chose to optimize for its own interests, like making more money."

The 60 Minutes interview

https://www.youtube.com/watch?v=_Lx5VmAdZSI



Lecture outlines

- Introduction to Ethics
- A procedure for ethical analysis
- What you can do if you find yourself involved in something you disagree with ethically at work.
- Code of conduct and code of ethics (ACM and BCS)
- **Activity - Seminar**

Part 1

Ethics Theories

Ethics

- ETHICS Defenion & Meaning

- 1 *(used with a singular or plural verb)* a system of moral principles:
the ethics of a culture.
- 2 *(used with a plural verb)* the rules of conduct recognized in respect to a particular class of human actions or a particular group, culture, etc.:
medical ethics;
Christian ethics.
- 3 *(used with a plural verb)* moral principles, as of an individual:
His ethics forbade betrayal of a confidence.
- 4 *(used with a singular verb)* that branch of philosophy dealing with values relating to human conduct, with respect to the rightness and wrongness of certain actions and to the goodness and badness of the motives and ends of such actions.: Compare [axiological ethics](#), [deontological ethics](#).



Categories of ethics

- **Descriptive Ethics:** Looks at what people actually do and what moral beliefs they follow (what do people do in real life?).
- **Metaethics:** Looks at the meaning of moral terms and ideas (what is morality really about?).
- **Applied Ethics:** Uses ethical principles to address practical issues in real situations such as medicine, technology, business, or law (how we should solve real moral problems).
- **Normative Ethics:** Establishes moral standards for actions (what should people do?).
 - **Virtue Ethics:**
 - **Consequentialism**
 - **Deontology**



Normative Ethics

Normative ethics focuses on how we determine whether an action is good. It asks not, “**what is good?**” but “**how do we decide what is good?**”



Useful to be able to reason using all of them

- helps you understand others' ethical arguments
- helps you see points you may have missed

Virtue ethics: Based on values

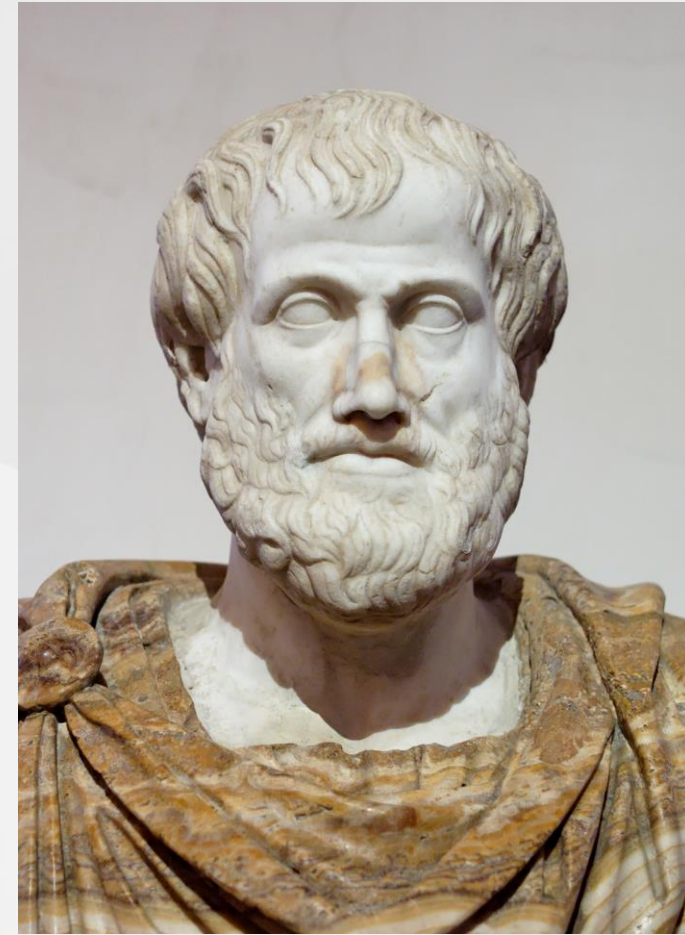
Consequentialist ethics: Based on outcomes

Deontic ethics: Based on rules



Virtue Ethics

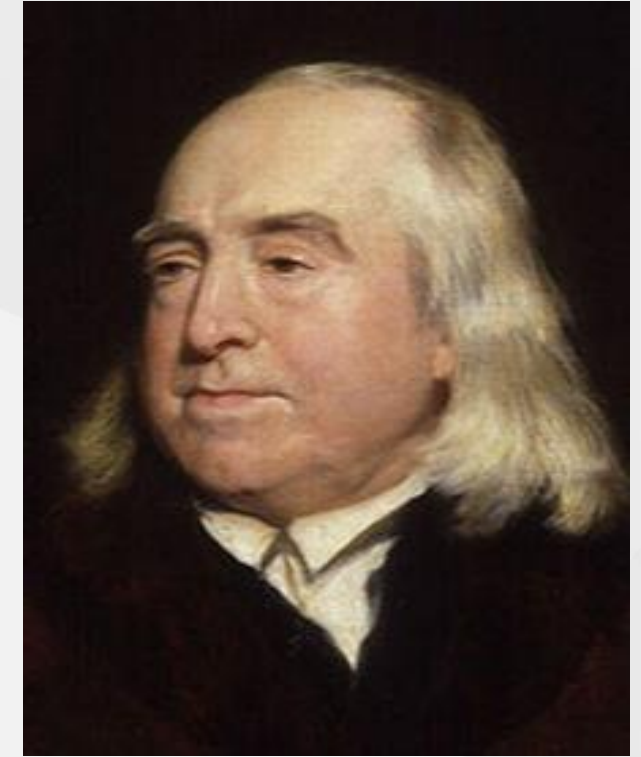
- Focuses on acting in line with positive moral values
- A behaviour is considered good if it reflects these values
- Emphasises developing a strong moral character through good habits
- **Aristotle (384–322 BC)** taught that virtue comes from practising good actions until they become part of who we are



[Aristotle \[Wikipedia\]](#)

Consequentialist Ethics

- Choose actions that lead to **the best overall outcomes**
- Based on **Utilitarianism**
- **Jeremy Bentham (1748–1832)**
- Goal: “**the greatest good for the greatest number**”
- Consider the impact on **all** stakeholders, not just yourself



[Jeremy Bentham \[Wikipedia\]](#)

Deontic Ethics

- **Focuses on following moral rules and duties**
- An action is right if it follows the **rules and principles**, and wrong if it does not
- **Consequences do not define morality; rules do**
- **Immanuel Kant (1724–1804)** developed this approach, teaching that moral duty comes from **within us**



Immanuel Kant
[Wikipedia]

Professional codes

- BCS (UK)
<https://www.bcs.org/media/2211/bcs-code-of-conduct.pdf>
- ACM (US)
<https://www.acm.org/binaries/content/assets/about/acm-code-of-ethics-and-professional-conduct.pdf>
- East Asia Academies of Engineering (China, Japan, Korea)
<http://ethics.iit.edu/ecodes/node/5076>
- EU ethics guidelines for Trustworthy AI
<https://ec.europa.eu/digital-single-market/en/news/ethics-guidelines-trustworthy-ai>

Some values in professional codes

- In all three professional codes (BCS, ACM, East Asia):
 - Public wellbeing, health and safety
 - Honesty, transparency
 - Non-discrimination/equality, inclusion
 - Accountability
 - Environmental sustainability
- In both the Computer Science codes (BCS & ACM):
 - Privacy, confidentiality
- EU AI guidelines: also Human agency & oversight

Case Study – Amazon Antitrust Paradox



↑ Rapid growth & market dominance

💰 Aggressive pricing & tactics



Harming Competition?

- Crushing smaller businesses



Hurting Consumers?

- Less choice, low wages



Power & Control?

- Too much influence?

“The company uses its size to stifle competition and exploit customers.”

– Lina Khan

Case study – amazon antitrust paradox

- 1. Use of data (2020):** Amazon was criticised for using customer and seller data to guide its business decisions and advertising.
- 2. Competing with sellers:** Amazon launched its own branded products that competed with successful third-party sellers on the platform.
- 3. Regulatory attention:** The criticism was set out in a paper by Lina Khan, who later worked on the US House investigation into Amazon and is now Chair of the Federal Trade Commission.
- 4. Amazon's position:** Amazon denied any wrongdoing and stated that there is no conflict of interest on its platform.

[Amazon's antitrust Paradox - Lina Khan, The Yale law journal.](#)

Stakeholders?

- Amazon sellers
- Amazon employees
- Amazon customers
- Amazon shareholders
- Other online retailers
- Online shoppers
- (more?)



Was it OK?

Why / why not?

(Outcomes – Rules – Values)

Some arguments against Amazon

Outcomes

- Unfair advantage over sellers
- Uses access to sellers' data

Rules

- Sellers take risks
- Amazon enters only after success

Values

- Lack of transparency
- Power imbalance

Some arguments for amazon

Outcomes

- More jobs
- Lower prices, more choice

Rules

- Sellers benefit from Amazon's platform
- Shared commercial success

Values

- Efficient delivery and storage
- Sustainability claims



What can you do?

Question:

What can you do if you find yourself involved in something unethical at work, as an employee or as an employer?

What can you do?

#1: don't get into this situation in the first place

- Due diligence in job applications
- May be possible to evade problem
- Keep your compass: have more than one community
- Routinely think about ethical issues of your work

What can you do?

#2: talk about it within the team/organization

- Get allies
- Look for corporate rules / outcomes / values supporting your case
- Be factual
- Suggest solutions if you can
- If low-key informal approach isn't enough, use company reporting procedures where they exist

What can you do?

#3: Public whistleblowing (e.g. talk to journalist)

For heroes/heroines.

Only try this after #2 has failed.

- Read this first: <https://www.gov.uk/whistleblowing>
- Support organizations: protect-advice.org.uk, whistleblowers.org
- NDAs invalid in some circumstances: check with a lawyer



What can you do as an employer

- Encourage open discussion and psychological safety
- Put a clear reporting process in place

UK govt guidance

<https://www.gov.uk/government/publications/whistleblowing-guidance-and-code-of-practice-for-employers>

Part 2

Professional code of ethics

Professionalism



Profession (From Oxford dictionary):
A type of job that needs special training or skill, especially one that needs a high level of education.

Regulations of a profession

Role of a Professional Body

- **Education and entry standards:** Sets required education and training.
- **Ethical standards and rules:** Defines expected professional behaviour.

Purpose of Professional Regulation

- **Protect the public and clients:** Client interests come first.
- **Independence with oversight:** Professions set standards; government safeguards the public interest

What is a professional code of conduct/ethics?

“The Code is designed to inspire and guide the ethical conduct of all computing professionals, including current and aspiring practitioners, instructors, students, influencers, and anyone who uses computing technology in an impactful way.”

<https://ethics.acm.org/>

What is a professional code of conduct/ethics?

- **Set by professional associations**
Organisations such as IEEE, ACM, and BCS set entry requirements.
- **Membership comes with responsibility**
Members accept a duty to meet professional standards of behaviour.
- **Written in a code of conduct**
The code explains expected behaviour and the core principles of the profession.

Computing in the UK

BCS – The Chartered Institute for IT

The professional body for computing and IT professionals in the UK.

Professional qualifications:

- Chartered IT Professional (CITP)
- Chartered Scientist (CSci)
- Chartered Engineer (CEng)

<http://www.bcs.org/>

BCS Code of Conduct

- **British Computer Society**
- The **BCS code of conduct** states that IT professionals should carry out their professional responsibilities with due care and diligence in accordance with the Relevant Authority's requirements whilst exercising your professional judgement at all times.

“Have due regard for public health, privacy, security and wellbeing of others and the environment” (BCS, Code of Conduct, 2011).

- <https://www.bcs.org/membership/become-a-member/bcs-code-of-conduct/>
- See the article: https://medium.com/@dadu_64294/the-british-computer-society-bcs-and-the-code-of-conduct-4efa47105a16



Who are IEEE and ACM?

- **IEEE Institute of Electrical and Electronics Engineers**

- The world's largest professional society, supported by a global community of engineers, professionals, and volunteers.
- https://www.ieee.org/content/dam/ieee.org/ieee/web/org/about/ieee_code_of_conduct.pdf

- **ACM Association of computing Machinery**

- Another professional society focused on computing and computer science, including areas such as data analytics, programming, data mining, web technologies, and software development.
- <https://ethics.acm.org/code-of-ethics/>

Code of Ethics vs. Code of Conduct

“A Code of Ethics governs decision-making, and a Code of Conduct govern actions. They both represent two common ways that companies self-regulate. They are often associated with large companies and provide direction to employees and establish a public image of good behavior, both of which benefit businesses of any size.”

Source: [Code of Ethics and Code of Conduct Difference: A Comparative Analysis](#)



ACM code of Ethics and Professional Conduct

A full explanation of each of these is found here: <https://www.acm.org/code-of-ethics>

1. GENERAL ETHICAL PRINCIPLES

- 1.1 Contribute to **society and to human well-being**, acknowledging that all people are stakeholders in computing.
- 1.2 Avoid harm.
- 1.3 Be honest and trustworthy.
- 1.4 Be **fair** and take action **not to discriminate**.
- 1.5 Respect the work required to produce new ideas, inventions, creative works, and computing artifacts.
- 1.6 Respect privacy.
- 1.7 Honor confidentiality.

ACM code of Ethics and Professional Conduct

2. PROFESSIONAL RESPONSIBILITIES

- 2.1 Strive to **achieve high quality** in both the **processes** and **products** of professional work.
- 2.2 Maintain high standards of professional competence, conduct, and ethical practice.
- 2.3 Know and **respect existing rules** pertaining to professional work.
- 2.4 Accept and provide appropriate professional review.
- 2.5 Give comprehensive and thorough evaluations of computer systems and their **impacts**, including analysis of possible **risks**.
- 2.6 **Perform work only in areas of competence.**
- 2.7 Foster **public awareness** and **understanding of computing, related technologies, and their consequences.**
- 2.8 **Access computing and communication resources** only when **authorized** or when compelled by the public good.
- 2.9 Design and implement systems that are **robustly and usably secure.**

ACM code of Ethics and Professional Conduct

3. PROFESSIONAL LEADERSHIP PRINCIPLES

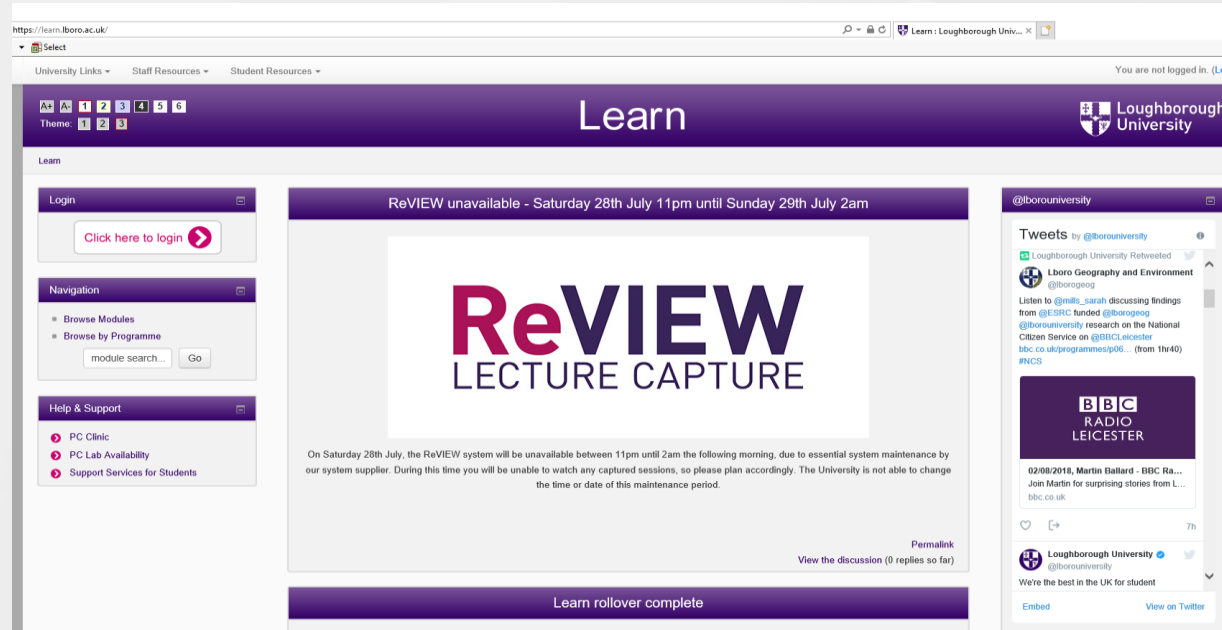
- 3.1 Ensure that the **public good is the central concern** during all professional computing work.
- 3.2 Articulate, encourage acceptance of, and evaluate fulfillment of social responsibilities by members of the organization or group.
- 3.3 Manage personnel and resources to **enhance the quality of working life**.
- 3.4 Articulate, apply, and support policies and processes that reflect the principles of the Code.
- 3.5 **Create opportunities** for members of the organization or group to grow as professionals.
- 3.6 Use care when modifying or retiring systems.
- 3.7 Recognize and take special care of systems that become integrated into the infrastructure of society.

4. COMPLIANCE WITH THE CODE

- 4.1 Uphold, promote, and respect the principles of the Code.
- 4.2 Treat violations of the Code as inconsistent with membership in the ACM.

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